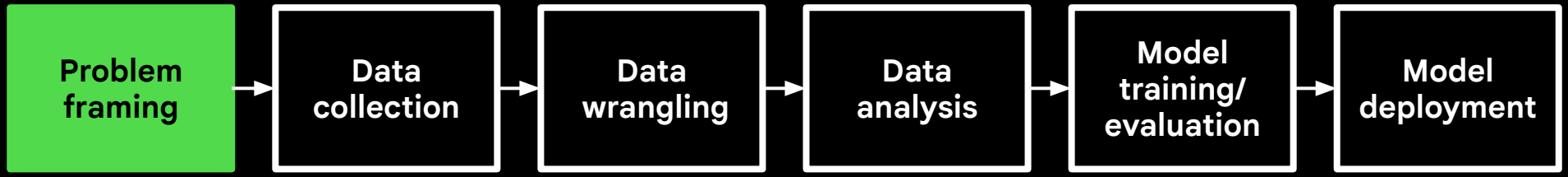


Photo by Google DeepMind on Unsplash

Machine learning project lifecycle



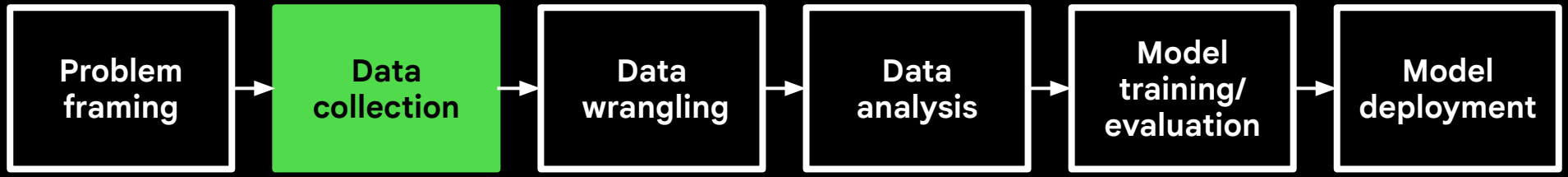
Express the goal clearly and concisely

- ❑ What is the main goal ? In which context? *(SMART)*
- ❑ What value does it add? (cost-benefit analysis, include maintenance)
- ❑ Can we solve it without ML? Is it feasible?
- ❑ Do we have enough quality data? (correct, representative, predictive power)

Express the goal technically

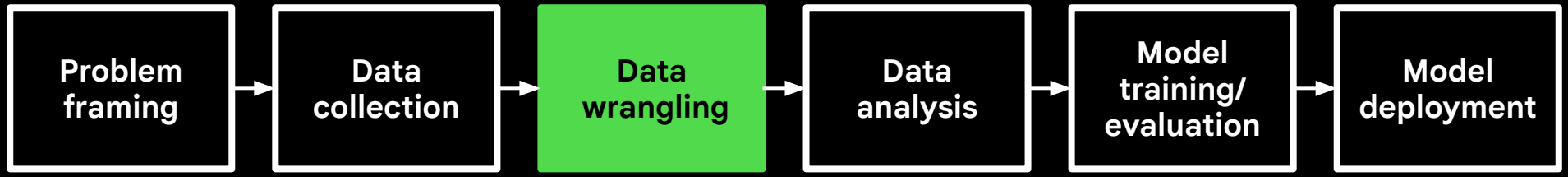
- ❑ What is the model's goal (measurable)? What is the success/failure metric?
- ❑ What are the Input and Output of the model?
- ❑ What is the performance measure?
- ❑ What are the non-ML baselines?

Machine learning project lifecycle



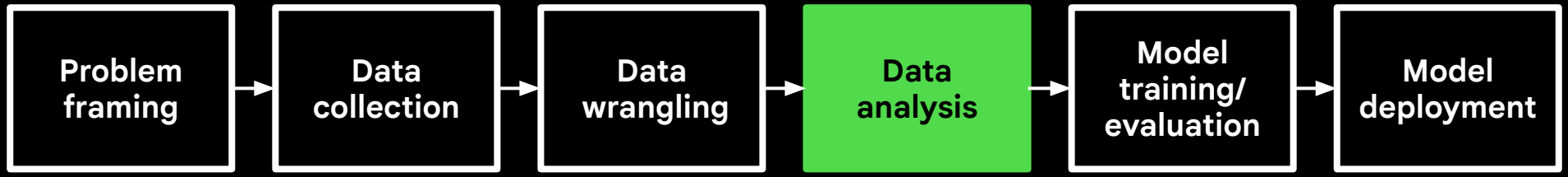
- ❑ Select the dataset size and the sampling strategy
- ❑ Identify feature and label sources (actual vs proxy labels)
- ❑ Measure data quality (noise, label or feature error, missing values, predictive power)
- ❑ Make sure data is representative of production use case (beware sampling bias)
- ❑ Split the data into train, validation , test (beware random splitting, fix random seed, beware data leakage, train older than test)

Machine learning project lifecycle



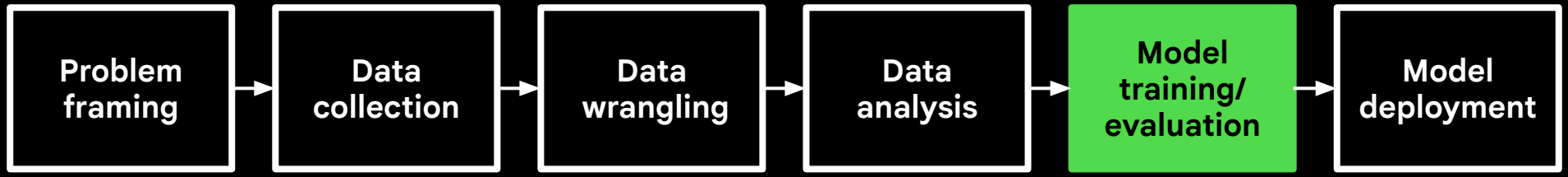
- ❑ Data transformation (convert non-numeric features to numeric, resize input to fixed size)
  - Numeric: Normalization (range scaling, log-scaling, clipping, z-score or standardization), binning (equally-spaced, quantile-based)
  - Categorical: one-hot encoding, tokenization
- ❑ Transform within a pipeline (beware data leakage)
- ❑ Data cleaning (missing values, imputation)
- ❑ Feature engineering (determining which features are important for training and creating them from raw data)

Machine learning project lifecycle



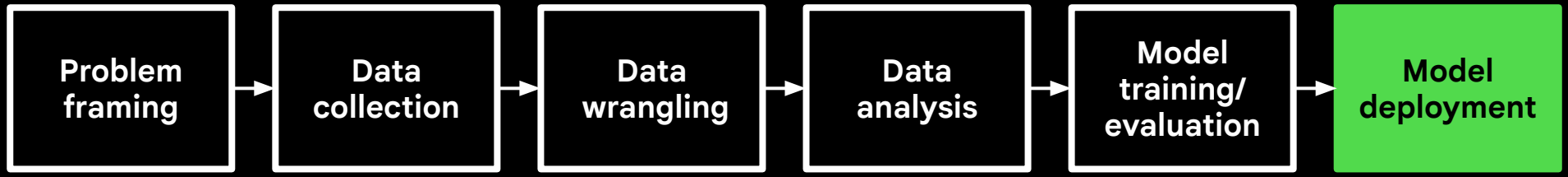
- ❑ Look at data distributions (summary statistics, histogram, density, CDF)
- ❑ (Periodically) validate the data (expected behavior with data schema; consistency over time; consider context, source and sampling strategy)
- ❑ Identify and decide what to do with outliers
- ❑ Handle noise (confidence interval, hypothesis testing)
- ❑ Look at individual examples
- ❑ Group the data (look from different subgroups perspective)
- ❑ Exploratory data analysis (graphs)
- ❑ Make hypothesis and look for evidence (scientific method)
- ❑ Remember correlation != causation

Machine learning project lifecycle



- ❑ Establish strong baselines
- ❑ Make sure (data) pipeline is correct before training (always validate data quality)
- ❑ Start with a simple model (then incrementally add complexity, train on small data first)
- ❑ Train on **training data**, select models and tune hyperparameters on **validation data** and test once on **test data** (update validation and test data)
- ❑ Debug based on loss and metric:
  - Debug data (validate input data with tests: correct, representative, predictive power; splits; preprocessing; numerical overflow)
  - Debug model: **Overfitting** (reduce model capacity, regularization, more data, train-test same distribution); **Underfitting** (increase model capacity, reduce regularization, feature engineering); Hyperparameter tuning; Feature selection (correlation with labels, performance on validation set)
- ❑ Document everything (especially failures)

Machine learning project lifecycle



- ❑ Periodically train model on new data
- ❑ Treat data and model as code (version control)
- ❑ Test each component of the pipeline (input data, data transformations, model updates, serving infrastructure)
- ❑ Integration test of end-to-end pipeline (when introducing new models or training on new data)
- ❑ Track training-serving skew (data schema skew, features skew, beware feedback loops, update the model on new data)
- ❑ Monitor model performance and efficiency (regression testing, checkpointing)


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